

CITYTWIN · DESIGN DECISION RECORD

CityTwin recommendations accepted

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DOCUMENT CONTROL

Document ID	CTW-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/citytwin		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record the recommendations accepted by Amish on 2026-09-25, what changed in the repo and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item below that carried a recommendation is decided by Amish, 2026-09-25: go with recommendation. Items with no recommendation stay "Proposed, awaiting Amish".

■ Context

On 2026-09-25 Amish wrote, in chat: "i accept all your recommendations, go with them across all repos." Before that, CTW-DDR-001 (D1 to D11) recorded this repo's TRL 2 review choices as adopted for TRL 3 work and open for his review, and the TRL 3 review note (docs/REVIEW.md , session 2026-09-25: TRL 3) listed six further items with options and a recommendation. This record lists every item now decided, what changed in the repo because of it, and what remains open. Where a recommendation offered several options, the recommended option is the decision. Work that belongs to TRL 4 (build, test, measurement, purchasing, firmware beyond a sketch) is decided but on hold, because TRL 4 is on hold by Amish's instruction.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
D1	Budget scope (CTW-DDR-001)	Option (b): the budget covers the kiosk; the TwinKit gateway is costed in TwinKit	Status wording only; R17 stays a reported figure
D2	Display	Color e-paper for the street kiosk, indoor LCD variant	Status wording; the indoor LCD variant is named in R10 and remains unmodeled at TRL 3
D3	Climate range	Sheltered or indoor pilot, heater study at TRL 3	Status wording (range narrowed by N5 below)
D4 to D10	Gateway in the cabinet; mains power with a solar variant; 5 per hour threshold and two years of raw data; CC BY 4.0; buttons; 46-node reference neighborhood; one-way publishing with privacy rules at the gateway	As recommended	Status wording in CTW-DDR-001 v0.2, CTW-PRB-001 v0.4, CTW-PRC-001 v0.4 and CTW-REQ-001 v0.4
D11	Pull paths for PotholeLog and DockHub	The gateway pulls published PotholeLog segment files and DockHub hourly aggregates by outbound HTTPS	Status wording; format agreement listed as a cross-repo action in <code>docs/REVIEW.md</code>
N1	Budget figure	<code>budget_usd</code> raised to the recommended \$750	<code>project.yaml</code> <code>budget_usd</code> \$600 to \$750; R16 now met on paper for the pilot kiosk (\$739.00, \$11.00 under); README budget line updated
N2	R10 button response	Option (a) for the street kiosk and (b) indoors: press acknowledged within 0.5 s by the button's LED ring and layer shown within 25 s; the indoor LCD variant keeps 5 s	R10 restated in CTW-REQ-001; firmware rule "LED ring lit from the press until the new image is drawn" added to CTW-PRC-001; CTW-CAL-001 [C2] checks both limits; R10 not met (19.8 s against 5 s) to met on paper (30 ms and 19.8 s against 0.5 s and 25 s)
N3	R18 outage ride-through	Option (b): ask TwinKit for a larger pack in cabinet installs	CTW-CAL-001 [I1] now sizes the pack: at least 1.84 Ah against TwinKit's 1.5 Ah. Request listed as a cross-repo action; TwinKit not edited. R18 stays not met until TwinKit changes its pack
N4	R20 street climate and the cabinet	A ventilated sun shield on the cabinet and a shaded screen orientation as site rules for any outdoor pilot; a panel heater only if the partner city has frost	New BOM item 19, cabinet sun shield (1.5 mm aluminum, 413 x 186 x 486 mm, 25 mm air gap, open bottom, top vents), \$30.00, in <code>cad/src/model.py</code> , STEP and STL, CTW-DWG-001 Rev P2 and all media. CTW-CAL-001 v0.2: cabinet rise in sun 18.7 K without the shield, 7.2 K with it; the pack stays under its 45 °C charge limit in full sun up to 37.8 °C air. Site rules added to CTW-PRC-001. R20 stays not met (panel 45.1 °C shaded at 45 °C air)

#	Item	Decision	What changed in the repo
N5	R12 pilot margin	Define the pilot range as 0 to 35 °C air	R12 target 0 to 40 °C to 0 to 35 °C; panel 35.1 °C (4.9 K margin), pack 39.4 °C (5.6 K margin); R12 at risk to met on paper
N6	Hood light (suggestion in the TRL 3 review note)	Dim the 2 W hood light to about 0.3 W, since 200 lx is enough	Controller rule in CTW-PRC-001; CTW-CAL-001 [D2, G1]: 219 lx at night, mains power 10.2 W to 9.2 W, 90 to 81 kWh a year

Table 2. Items still open.

#	Item	Status
O1	First partner city or neighborhood for co-design and a pilot site. No recommendation was made.	Proposed, awaiting Amish
Q1	Antenna height needed for the reference neighborhood with the gateway in the cabinet	Open question, no recommendation made
Q2	Languages and scripts on the kiosk, and how residents ask questions or report a fault	Open question, no recommendation made; for co-design with the partner city (O1)

■ Consequences

- `project.yaml`: `budget_usd` \$750. Pitch and problem unchanged: no rewording was recommended. `trl: 3` and `trl_target: 3` unchanged.
- Documents revised: CTW-PRB-001 v0.4, CTW-PRC-001 v0.4, CTW-REQ-001 v0.4, CTW-CAL-001 v0.2, CTW-DDR-001 v0.2; drawing CTW-DWG-001 Rev P2; `bom/bom.csv` (19 lines) and `bom/bom-notes.md`; `README.md`.
- Requirement status (CTW-CAL-001 v0.2): 2 not met (R18, R20), 1 at risk (R1), 10 met on paper, 7 met by design. Before: 4 not met, 2 at risk, 7 met on paper, 7 met by design.
- The sun shield adds 1.86 kg and 35 N of gust load: post stress 26.2 to 26.9 MPa, anchor tension 1.92 to 1.97 kN, mass 58.2 to 60.1 kg. All still met.
- Cross-repo actions (TwinKit pack, PotholeLog and DockHub pull formats, CrossSafe uplink) are listed in `docs/REVIEW.md`. No other repo was edited.
- On hold at TRL 4 by Amish's instruction: the panel heater (only for a frost site), the indoor LCD and solar variants beyond their mention, and every build, bench timing, climate chamber and outage test. Nothing in this record authorizes building or testing.